



Smart World

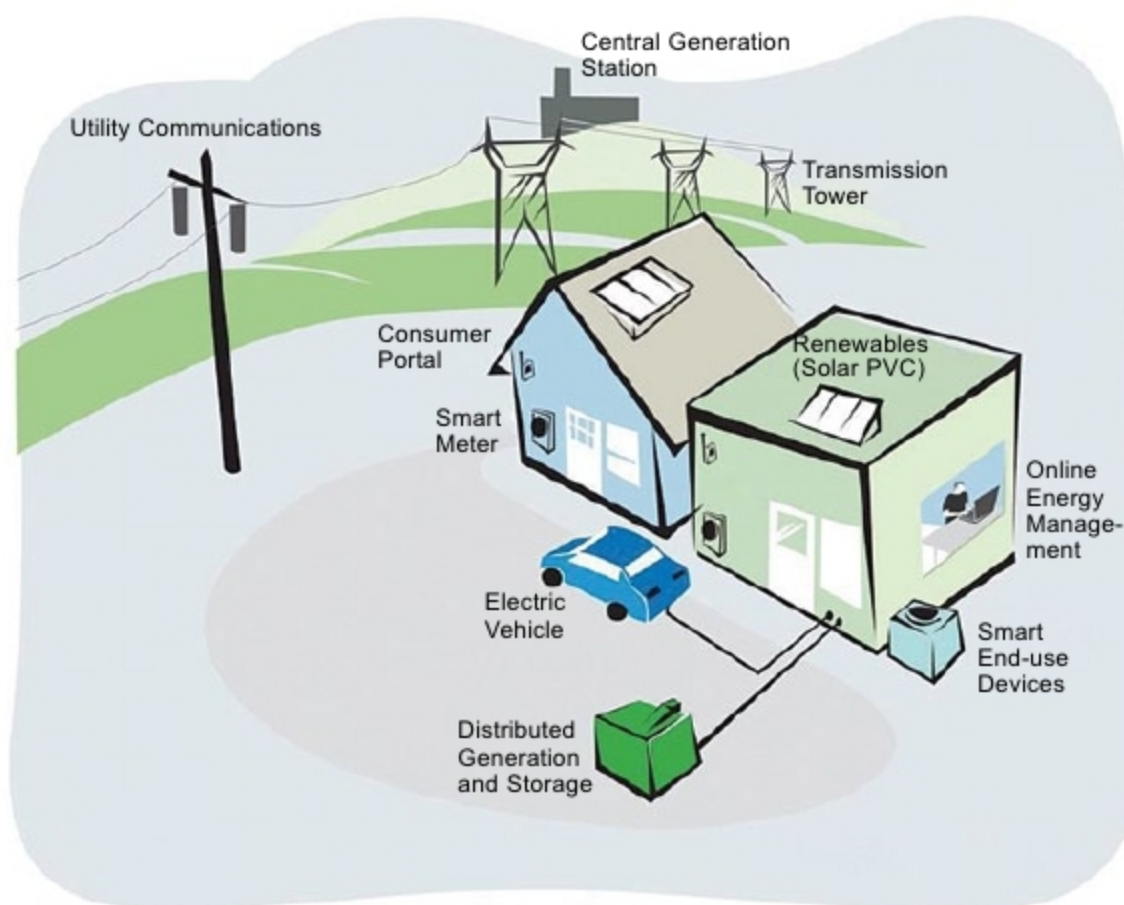
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1 SMART POWER SOLUTIONS FOR SMART CITIES

Sustainable generation of electricity is usually done using available renewable options such as solar rooftops, wind and biomass



(Credit: PortlandGeneral.com)

Most smart cities have the same amenities that vary a little depending on the geographical region. Without an uninterrupted supply of power, a city cannot be holistically smart or provide e-mobility for applications like smart poles, CCTV surveillance, smart parking lots, etc. One of the best possible ways to ensure no power outages is the use of a smart grid.

According to www.amarketresearchreport.com, the smart grid technology market is expected to register a compound annual growth rate (CAGR) of 23 per cent during 2019-2026. Explaining the importance of smart grid, Amruth Puttappa, founder, ThingsCloud, says, "A smart grid self-detects the concerns and heals the network on its own without any manual intervention. This makes

smart grid inverters more reliable and secure during natural disasters. A smart grid improves strength of the distributed energy generation system and enables the option for more solar rooftop systems and small wind turbines."

Prior to installation of the grid at a location, processes like load analysis, site and resource evaluation need to be carried out. Accordingly, financing options can be determined once location-specific modelling and construction is finalised. Transformers mounted on poles, underground feeder pillars and so on help in dealing with space constraints when it comes to outdoor applications.

EVs increase electricity demand, which adds stress to the local distribution system. Smart charging solutions, like re-charging overnight, can help minimise this. EVs could feed electricity back to the grid at times of high demand and contribute to stability of the charging grid. These could also act as distributed batteries for storage.

By upgrading the existing transmission and distribution system, reduced loss levels can be achieved. Puttappa explains, "If the source is closer to the consumer, transmission loss can be reduced. Utilisation of microgrids always ensures minimal transmission and distribution losses. We should start considering standalone solar sys-